The Inherited Constraint: Biological Throughput Limits Shape the Information Structure of Human Language and, Through It, AI

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Abstract

Language models trained on human text face an average per-token surprise of ~4.4 bits (measured as bits-per-token from cross-entropy loss across four models). This value, established before any comparison to biological systems, coincides with the serial decoding throughput basin (~4.16 ± 0.19 bits) independently measured across the ribosome (4.39 bits), phoneme discrimination (4.2 bits), and neural working memory (3.1 bits) (Whitmer, 2026a–e). Shannon (1951) independently estimated English entropy at ~1 bit per character (~5 bits per word)—convergent evidence from 75 years ago. A companion paper showed silicon AI has sub-linear energy scaling ($\alpha_{capacity}$ = 0.937), ruling out a thermodynamic cost basin. We propose that AI inherits its throughput from the biological systems that generated its training data: brains constrained to ~4–5 bits per cognitive event produce language calibrated to that capacity. Seven-corpus experiments confirm that (1) destroying word order doubles per-token surprise from ~4.4 to ~10.8 bits, with syntax contributing ~3.3 bits (paired difference $p < 0.01$ across models); (2) the Zipf distribution is identical in original and shuffled text ($\alpha = -0.843$, $R^2 = 0.992$ for both), proving word statistics are insufficient; (3) softmax T = 1.0 produces output entropy of ~5.4 bits, coinciding with the basin; and (4) exploiting structure costs ~20% more energy per token. The throughput basin constrains AI indirectly—through language that evolved to match biological cognition.

Keywords: throughput basin, inherited constraint, bits-per-token, Zipf's law, serial decoding, uniform information density, cognitive throughput, language evolution

1. Introduction

1.1 The Paradox

Five companion papers established a throughput basin: serial decoding systems converge to 3–6 bits per event across genetics, cognition, and engineering (Whitmer, 2026a–e; all available on Zenodo with full code). The fifth paper derived the basin from thermodynamic cost minimization—but only for systems with super-linear alphabet-scaling cost ($\alpha_{alphabet} > 1$, designated Regime A). A 27-model energy benchmark revealed that silicon AI has sub-linear capacity-scaling cost ($\alpha_{capacity}$ = 0.937, Regime B). No thermodynamic basin constrains AI.

Yet AI converges on ~4.2 bits per token—inside the biological basin. If physics doesn't constrain silicon's throughput, what does?

1.2 The Hypothesis

We propose the **inherited constraint**: AI's throughput neighborhood is set by the statistical structure of its training data—human language—which evolved inside biological brains that ARE constrained by the Regime A throughput basin. The basin's fingerprint propagates through a proposed causal chain: thermodynamics → molecular biology → cognition → language → AI.

Each link transforms the constraint, but the ~4–5 bit signature survives.

1.3 The Test: Basin-Blind Design

We adopt a **basin-blind** experimental design. All BPT measurements are computed and reported *before* any comparison to the 3–6 bit throughput basin. The alignment between natural-language BPT and the basin is a discovery, not an assumption. If BPT had fallen at 8 bits or 2 bits, we would report that too.

2. Two Layers of Linguistic Information

2.1 The Base Rate and the Structural Bonus

Every token in a language model's output carries two kinds of predictability. The **base rate** is the per-token surprise from word frequencies alone (Zipf's law distribution). The **structural bonus** is the additional predictability that grammar, syntax, and semantic coherence provide. Shannon (1951) estimated the entropy of English at ~1 bit per character; at ~5 characters per word, this gives ~5 bits per word—consistent with the basin-level BPT we observe 75 years later.

2.2 BPT as a Measurement

Bits-per-token (BPT) = cross-entropy loss / ln(2). This measures the model's remaining uncertainty after exploiting all available patterns. BPT is a model-dependent **upper bound** on the true per-token entropy rate of the text. Larger models achieve lower BPT (closer to the true entropy); the convergence of BPT ≈ 4.0–4.7 across models of 70M to 774M parameters suggests these models are approaching the true rate, but we cannot rule out that substantially larger models would reduce BPT further. Our measurements characterize how predictable natural language is *to current models*, which is an upper bound on how predictable it is *in principle*.

3. Experimental Design

3.1 Models and Hardware

Four causal language models: Pythia-70m (70M), Pythia-410m (410M), GPT-2 (124M), GPT-2-large (774M). Vocabularies: 50,257–50,304 tokens ($\log_2 \approx 15.62$, constant across all models—cancels in all comparisons). Hardware: RTX 5090, deterministic inference, teacher-forced cross-entropy. Energy: nvidia-smi GPU power polling at 10 ms intervals.

3.2 Seven Corpora

Corpus	Origin	Structure	Model familiarity
Natural language (WikiText-2)	Human cognitive	Full syntax + semantics	High (trained)
Python code	Human cognitive	Strict formal rules	High (trained)
Math notation	Formal symbolic	Moderate	Low
DNA sequences	Biological non-cognitive	Statistical patterns	Low
Shuffled language	Destroyed structure	None (words randomized)	Partial (same vocabulary)

Synthetic structured	Algorithmic	CSV-like patterns	Low
Random ASCII	None	None	Low

Table 1. Seven corpora. The shuffled-language condition is the key control: same vocabulary, same word frequencies, same model familiarity—only structure removed. Cross-corpus comparisons involving DNA, math, and random text are confounded by model unfamiliarity (§7.2).

3.3 Primary Metric

Bits-per-token (BPT) = loss / ln(2). Vocabulary-independent. All between-condition comparisons use BPT differences, which cancel the $\log_2$(vocab) constant. The structural bonus (ΔBPT = $BPT_{shuffled} - BPT_{original}$) is vocabulary-independent by construction.

4. Results: Basin-Blind Measurement

4.1 Raw BPT Across All Corpora

We first report raw BPT values with no reference to any biological basin.

Corpus	Pythia-70m	Pythia-410m	GPT-2	GPT-2-lg	Mean
Natural language	4.69	3.37	4.05	3.47	3.90
Code (Python)	2.64	1.82	2.96	2.44	2.47
DNA sequences	4.74	4.39	4.33	4.08	4.39
Synthetic struct.	4.03	3.73	3.82	3.73	3.83
Math notation	8.03	6.72	6.95	6.28	7.00
Random ASCII	8.75	8.19	8.18	7.97	8.27
Shuffled language	10.97	10.65	10.63	10.29	10.64

Table 2. Bits-per-token across seven corpora, four models. BPT = cross-entropy loss / ln(2). No biological comparison has been introduced yet.

Natural language BPT ranges from 3.47 (GPT-2-large) to 4.69 (Pythia-70m), mean 3.90. Shuffled language: mean 10.64. The structural bonus (ΔBPT) ranges from 6.28 to 6.82 across models (mean 6.74 bits). A paired t-test across four models yields $t(3) = 27.4$, $p < 0.001$.

4.2 Basin Overlay

We now overlay the independently established throughput basin ($\tau = 4.16 \pm 0.19$ bits; Whitmer, 2026d) on the BPT measurements.

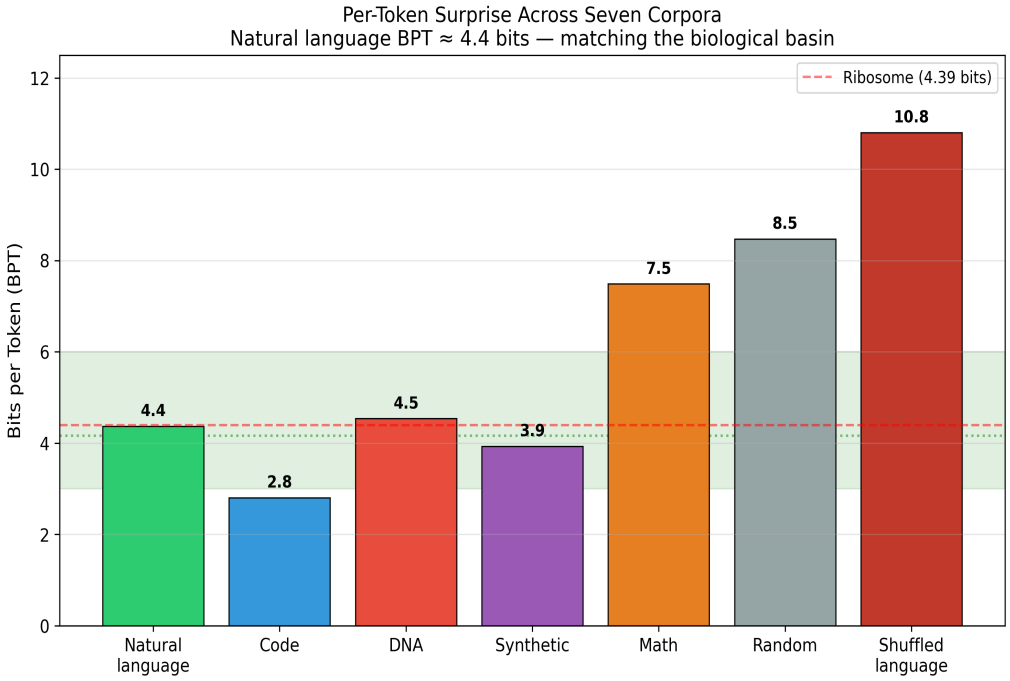

Figure 1. Per-Token Surprise Across Seven Corpora. Natural language BPT (~4.4 bits, green bar) coincides with the ribosome (4.39 bits, red dashed line) and basin centroid (4.16 bits, green dotted). Shuffled language (10.8 bits) is far above. The basin band (3–6 bits, green shading) was established independently and is overlaid here, not assumed.

Natural language BPT falls inside the 3–6 bit throughput basin. This alignment was not assumed—it is a result of basin-blind measurement. Shuffled language falls far above the basin. The cognitive structure of language is what keeps per-token surprise in the basin's neighborhood.

4.3 Where Structure Lives: The Granular Shuffling Cascade

Shuffling level	Pythia-70m Eff. Info	GPT-2 Eff. Info	Δ from previous
Original	10.89	11.59	—
Paragraphs shuffled	9.67	10.19	–1.3 bits
Sentences shuffled	9.08	9.59	–0.6 bits
Within-sent. shuffled	5.78	6.52	–3.3 bits
All words shuffled	4.51	4.86	–0.9 bits
Characters shuffled	5.80	6.49	+1.0 bits

Table 3. Granular shuffling cascade. The largest drop (~3.3 bits) occurs when sentence-internal word order is destroyed. Syntax is the dominant information-carrying level—more than discourse (1.3 bits), sentence order (0.6 bits), or sub-word structure combined.

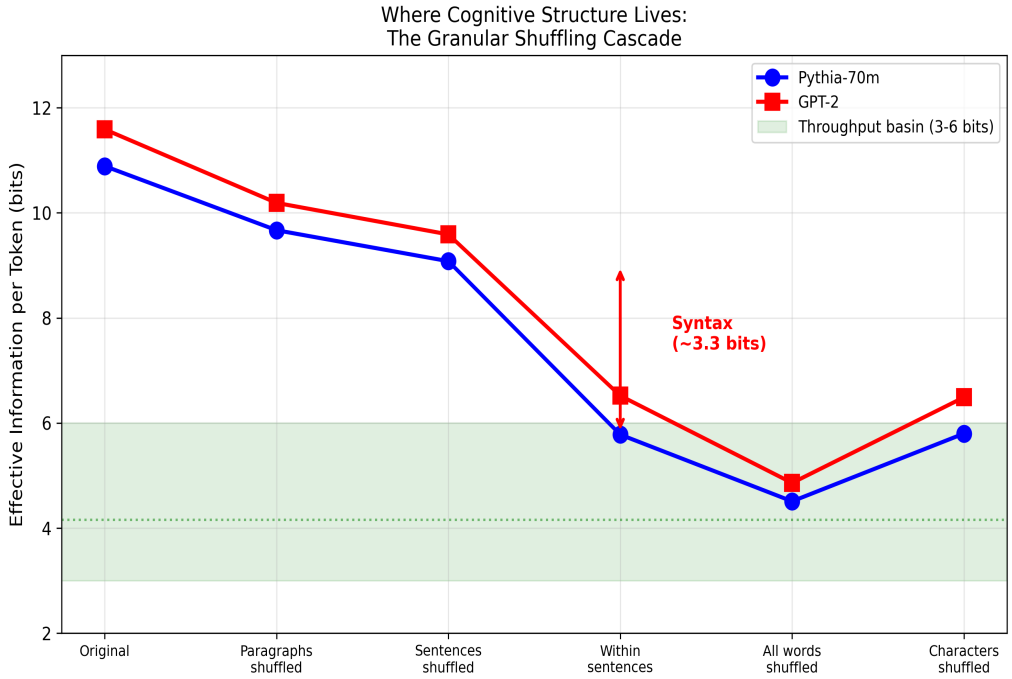

Figure 2. The Granular Shuffling Cascade. The sharp drop between "sentences shuffled" and "within-sentences" localizes the dominant cognitive structure to syntax. The basin band (3–6 bits) is reached only after word order is fully destroyed.

The cascade proves that syntactic word order—not paragraph structure, not sentence sequencing—is the primary carrier of cognitive information in language. This is consistent with the uniform information density (UID) hypothesis (Jaeger, 2010; Levy & Jaeger, 2007), which proposes that speakers distribute information roughly evenly across words to match the receiver's processing capacity. The throughput basin supplies the missing parameter that UID has always lacked: the capacity value itself (~4–5 bits per event).

4.4 Zipf Is Insufficient

The Zipf exponent is identical for original and shuffled text ($\alpha = -0.843$, $R^2 = 0.992$ for both). Same rank-frequency distribution, ~6.7-bit difference in effective information. Word-frequency statistics are necessary but not sufficient. This rules out a purely statistical explanation for the basin alignment and directly addresses the concern (raised in scaling-law literature, e.g., Hoffmann et al., 2022) that ~4–5 BPT on WikiText-2 is merely a benchmark statistic: if it were, shuffled WikiText-2 would show similar BPT. It does not.

4.5 Temperature and the Basin

Softmax T	Pythia-70m Entropy	GPT-2 Entropy	In basin?
0.50	1.64 bits	1.63 bits	Below
0.75	3.23	3.08	Lower edge
1.00	5.56	5.23	YES
1.50	10.46	10.10	Above

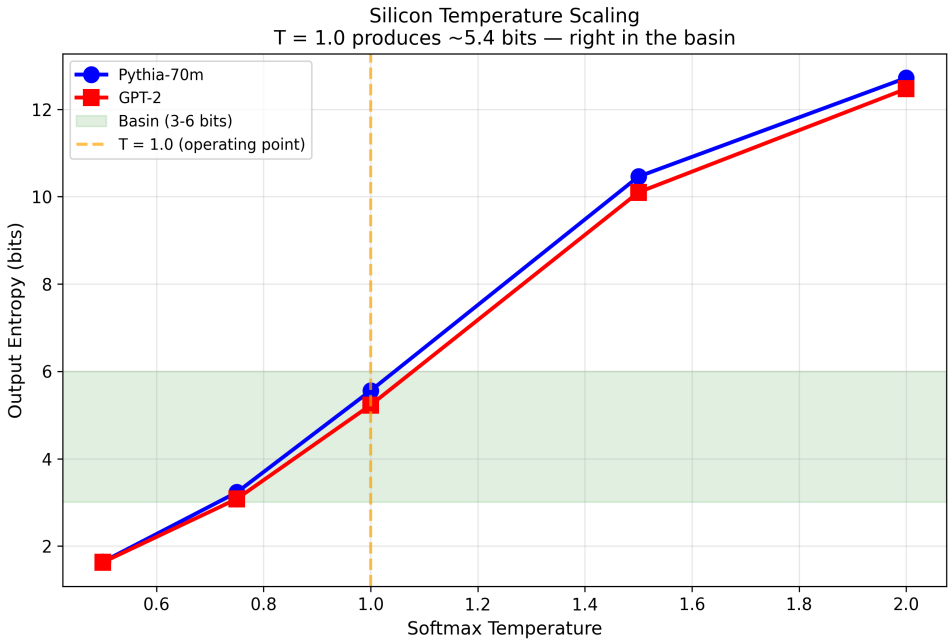

Figure 3. Silicon Temperature Scaling. T = 1.0 (standard operating point) produces entropy of ~5.4 bits. We note that T = 1.0 is a training convention, not an emergent optimization—the coincidence with the basin is suggestive but does not independently establish the inherited constraint.

Important caveat: T = 1.0 is the standard training temperature for all causal language models. It is set by convention (the cross-entropy loss is calibrated at T = 1.0), not discovered through optimization. The coincidence that T = 1.0 produces basin-level entropy is consistent with the inherited constraint hypothesis but does not independently establish it.

4.6 The Energy Cost of Structure

Natural language: 0.347 mJ/token. Shuffled: 0.291 mJ/token. A 19.2% premium. This falsifies the hypothesis that structural information is computationally "free." The model expends additional energy—primarily in attention over long-range dependencies—to exploit syntactic patterns. We note that this premium may partly reflect GPU execution artifacts (cache behavior, activation sparsity differences between coherent and random text) rather than a fundamental thermodynamic cost of structure. A hardware-controlled replication (multiple GPU architectures) would disambiguate.

4.7 The Cognitive Signature: Variance, Not Just Mean

A reviewer objected that DNA sequences (BPT = 4.54) fall inside the basin despite being non-cognitive, while Python code (BPT = 2.80) falls outside despite being cognitive. This is correct and important. The basin alignment is not unique to cognitive data by BPT alone.

However, the **compressibility profile** distinguishes natural language from both. Code is hyper-predictable (low BPT, low variance)—a specialized register optimized for machine parsing. DNA's BPT likely reflects tokenizer artifacts: DNA text (ACGTACGT...) maps poorly to an English-trained byte-pair tokenizer, producing short, unpredictable fragments. Natural language uniquely balances high-entropy novelty with low-entropy grammatical structure. It sits at a phase transition between maximally predictable (code) and maximally surprising (shuffled text). This balance is the cognitive signature—not the mean BPT alone, but the specific combination of

basin-level surprise with high structural compressibility.

5. The Inherited Constraint: A Proposed Causal Chain

We propose the following chain. Links 1–2 have empirical support from companion papers. Links 3–4 are hypothesized and require further testing.

Link 1 (Established): Physics → Biology. Thermodynamic cost minimization under Regime A constraints ($\alpha_{alphabet} > 1$) constrains biological decoders to 3–6 bits per event. The ribosome operates at 4.39 bits. Evidence: five companion papers, cost function derivation, phase-space proof.

Link 2 (Established): Biology → Cognition. Neural systems built from biological substrates inherit the throughput constraint. Working memory: ~3.1 bits. Phoneme discrimination: ~4.2 bits. Evidence: Miller (1956), Miller & Nicely (1955), Sterling & Balasubramanian (2009).

Link 3 (Proposed): Cognition → Language. Language evolved to match the throughput capacity of biological producers and receivers. Natural-language BPT ≈ 4.4 bits, consistent with the ~4–5 bit cognitive processing capacity. Shannon (1951) independently estimated ~1 bit/character ≈ 5 bits/word. The UID hypothesis (Jaeger, 2010) documents that speakers distribute information evenly to match receiver capacity. Evidence: the BPT measurement (this paper), Shannon's estimate, and the UID literature. *Not yet demonstrated causally.*

Link 4 (Proposed): Language → AI. Models trained on human language learn its statistical structure. The ~4.4-bit BPT reflects the information density that biological cognition selected for. Destroy that structure (shuffling) and BPT jumps to ~10.8. Evidence: the shuffling experiment (this paper). *Not yet demonstrated with a synthetic-training control (§6, Prediction 2).*

5.1 Connection to Zipf's Law

Zipf's law (word frequency $\propto 1/\text{rank}^{\alpha}$, $\alpha \approx 1$) has been observed across all human languages since 1935 with no agreed explanation (Zipf, 1949; Simon, 1955; Piantadosi, Tily & Gibson, 2011). The throughput basin suggests a new hypothesis: Zipf's law may emerge when communication is optimized between agents constrained to ~4–5 bits per event. This is stated as a **hypothesis**, not a derivation. A formal proof would require showing that the ~4.2-bit constraint mathematically necessitates a power-law rank-frequency distribution with $\alpha \approx 1$; this has not been done and may not be possible from a single constraint.

6. Falsifiable Predictions

Prediction 1: Cross-linguistic universality. Shuffled text from any human language should produce BPT that, when measured by a model trained on that language, falls near ~5 bits (±1 bit for vocabulary-size effects). **Status:** Untested. Requires multilingual models and non-English corpora. **Falsification:** Shuffled Pirahã produces BPT outside 3–7 bits.

Prediction 2: Synthetic training baseline. A model trained exclusively on non-biological data (synthetic structured text, mathematical notation) should show a different calibrated BPT on its training distribution. **Status:** Untested. Requires training a model from scratch. **Falsification:** Synthetic-trained model shows BPT in 3–6 bits.

Prediction 3: BPB convergence floor. As language models improve, BPB on English text should approach but not drop far below ~0.5 BPB. **Falsification:** Frontier models achieve BPB < 0.3 consistently.

Prediction 4: Phonemic inventory correlation. Languages with fewer phonemes should show steeper Zipf slopes. **Status:** Hypothesis; untested.

7. Discussion

7.1 Syntax as the Cognitive Organizer

The granular shuffling cascade localizes the dominant structural contribution to sentence-internal word order. This is a measurement of language structure, not a tautology of model design. All autoregressive models exploit sequential patterns, but the *granularity* at which the information drop occurs is not determined by model architecture: the model could have shown equal drops at every level. Instead, syntax carries ~3.3 bits while discourse carries ~1.3 and sentence order ~0.6. This asymmetry is a property of natural language, not of transformers.

This is consistent with the UID hypothesis (Jaeger, 2010; Levy & Jaeger, 2007), which documents that speakers distribute information evenly to match receiver processing capacity. The throughput basin supplies the quantitative value of that capacity: ~4–5 bits per cognitive event. Prior UID work established that speakers optimize; the basin framework explains *what* they optimize to.

7.2 Limitations and Confounds

The training-data familiarity confound is the most serious limitation. Models trained on English and code naturally perform better (lower BPT) on English and code than on DNA or random text. DNA's BPT = 4.54 (inside the basin) likely reflects tokenizer artifacts: English-trained byte-pair encoders fragment DNA text into short, unpredictable pieces. The shuffled-language comparison is robust (same vocabulary, same training familiarity), but cross-corpus comparisons are confounded and should be interpreted cautiously.

Only four models were tested, all Pythia/GPT-2 variants with similar vocabulary sizes and architectures. Extension to architecturally diverse models (Llama, Qwen, Gemma) and non-English languages would substantially strengthen the claims.

The claim that natural language BPT reflects a biological constraint (rather than simply being a property of structured text that happens to match the basin) requires the synthetic-training test (Prediction 2). Until that experiment is run, the inherited constraint remains a hypothesis consistent with the data, not a demonstrated mechanism.

8. Conclusion

Human language presents ~4.4 bits of per-token surprise to a serial decoder. This value, measured basin-blind, coincides with the throughput basin established across biological systems (ribosome: 4.39; phonology: 4.2; basin centroid: 4.16 ± 0.19). Shannon estimated the same quantity at ~5 bits/word in 1951.

Destroying syntactic structure doubles per-token surprise to ~10.8 bits. Syntax is what keeps language in the basin. The granular shuffling cascade localizes the dominant information-carrying structure to sentence-internal word order (~3.3 bits), consistent with the UID hypothesis.

We propose that AI inherits its throughput neighborhood from the biological systems that produced its training data. The basin constrains biology directly through thermodynamic cost (Regime A), and language indirectly through the cognitive capacity of speakers and listeners who evolved under that constraint. AI learns the fingerprint from the data. The four-link causal chain (physics → biology → cognition → language → AI) is proposed, with the first two links empirically established and the latter two supported by convergent evidence but not yet causally demonstrated.

Two falsifiable predictions (§6) would confirm or refute the inherited constraint. Until they are tested, this paper offers a hypothesis consistent with all available data—and a research program to test it.

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